

Luca Palmieri

**Notes for the course of "Fibers Optics"**

# **PRODUCTION OF OPTICAL FIBERS**

(Academic year 2019/2020)

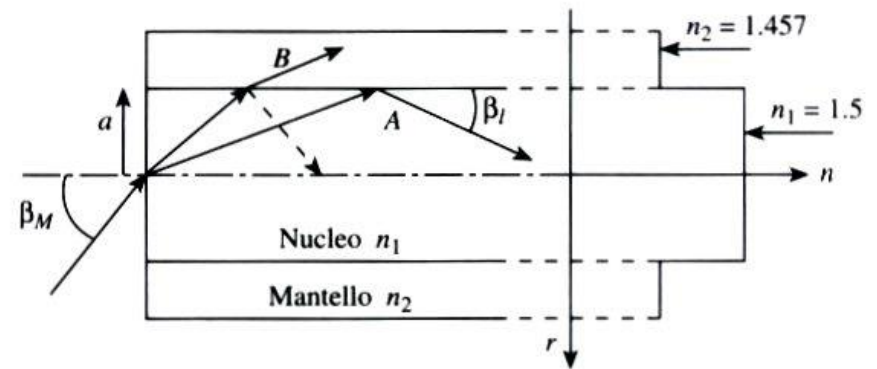
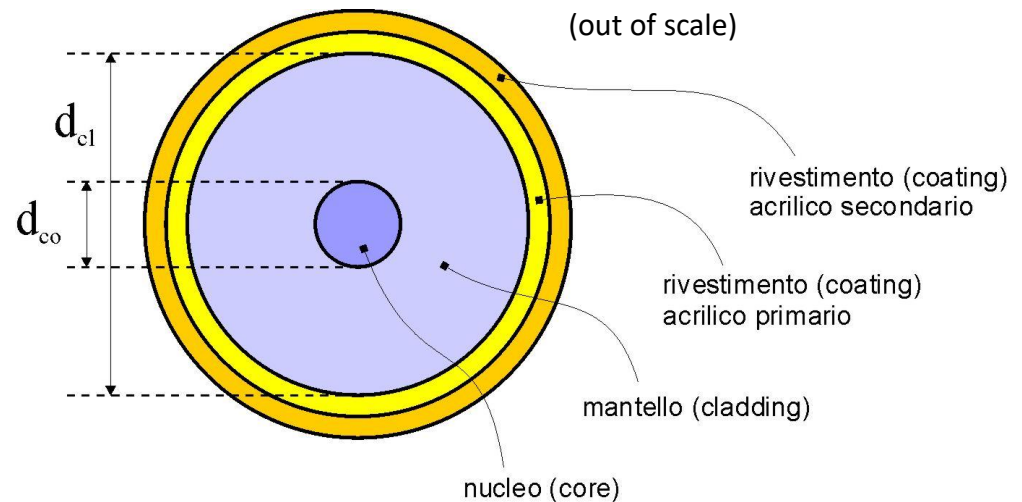
# Introduction

Optical fibers are made of a central core, surrounded by an external cladding. The refractive index of the core is slightly higher than that of the cladding:

$$n_{co} \approx 1.466 \dots \quad d_{co} \approx 6 \div 65 \mu m$$

$$n_{cl} \approx 1.444 \dots \quad d_{cl} \approx 125 \mu m$$

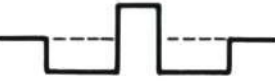
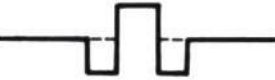
During propagation, light is mainly confined in the core. In a first order approximation, the confinement mechanism is total internal reflection. A more accurate description of propagation relies on modes.



**TE<sub>0,p</sub>   TM<sub>0,p</sub>   HE<sub>m,p</sub>**  
**EH<sub>m,p</sub>**

# Introduction

The propagation properties of an optical fiber are defined by the **refractive index profile**.

| INDEX PROFILE                                                                        | FIBER TYPE                       | MAIN FEATURES                                       |
|--------------------------------------------------------------------------------------|----------------------------------|-----------------------------------------------------|
|    | STEP - INDEX<br>MATCHED CLADDING | OPTIMIZED AT 1300nm                                 |
|    | DEPRESSED<br>CLADDING            | LOW ATTENUATION<br>LOW SENSITIVITY<br>TO BENDING    |
|    | W FIBER                          | DISPERSION<br>FLATTENED                             |
|    | PURE SILICA<br>CORE              | VERY LOW<br>ATTENUATION                             |
|    | TRIANGULAR CORE                  | DISPERSION SHIFTED<br>( OPTIMIZED AT 1550nm )       |
|   | SEGMENTED CORE                   | DISPERSION SHIFTED<br>LOW SENSITIVITY<br>TO BENDING |
|  | MULTIPLE CLADDING                | DISPERSION<br>MODIFIED<br>( SHIFTED OR FLATTENED )  |

( - - - - - : SiO<sub>2</sub> level )

# Introduction

Optical fibers can be made of silica ( $\text{SiO}_2$ ) or with some polymers.

**Silica optical fibers** are the one used in all high-capacity telecommunication systems, because of their superior optical properties.

**Polymer optical fibers (POF)** have much higher attenuation, but are cheaper to produce and install. Their use is limited to small networks, such as those in buildings or in cars and airplanes.

These notes are about silica optical fibers.



# Production techniques

The production of an optical fiber cable consists of 3 main phases.

- 1. Preform production.** The preform is a silica cylinder that reproduces in a larger scale the targeted refractive index profile. They are about 1 m long and from few up to few tens of centimeters in diameter. A single preform can yield up to several hundreds of kilometers of optical fiber.
- 2. Fiber drawing.** In this phase the preform is heated up to its viscous state, and drawn into a fiber. The process takes place in special drawing tower, several meters tall. During this phase the fiber is also coated with an acrylic layer (coating) to protect it from accidental scratches. The drawing speed can be as high as several tens of meters per seconds.
- 3. Cabling.** The drawing process produce a coated optical fiber with a standard outer diameter of 0.250 mm. Despite this "bare" fiber is stiffer than a steel wire with the same diameter, it needs further mechanical protection prior to being installed. The structure of the cable varies largely depending on the final application.

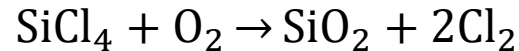
# The preform

There are 2 main technics to produce the preform.

1. By deposition of silica obtained by synthesis from a vapor phase. This technic belongs to the class of processes known as **chemical vapour deposition (CVD)** and is by far the most used system. It enable the production of silica with an extremely high level of purity and, at the same time, a tight control on dopants concentration.
2. By production of silica from a wet phase. These are the processes known as **sol-gel**, in which silica and other constituents are first held together by a gel matrix and then sintered in the final preform. It is a very effective approach to obtain silica with high levels of dopants or strongly heterogeneous. It also has a higher yield, compared to CVD techniques; however, the final quality of the silica is lower. They are sometimes used to make the outer layers of the cladding.

# Chemical vapor deposition

CVD is a technique used to produce material of the uttermost purity; it is widely employed, for example, by the semiconductor industry. CVD of silica is based on the following chemical reaction



through which vapors of silicon tetrachloride (liquid between -70°C and 57.6°C) are oxidized at high temperatures (1200÷1600°C) to produce silica and molecular chlorine. Silica precipitates as a soot to form the preform. The reaction is irreversible.

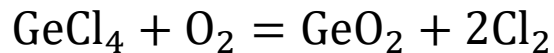
One of the advantages of CVD is that the attained silica is more pure than the original reactants (whose starting purity is already quite high, given their large use in the semiconductor industry). This is possible basically because silicon tetrachloride evaporates at much lower temperatures compared with the typical contaminants.

The refractive index of pure silica is 1.444 at 1550 nm.

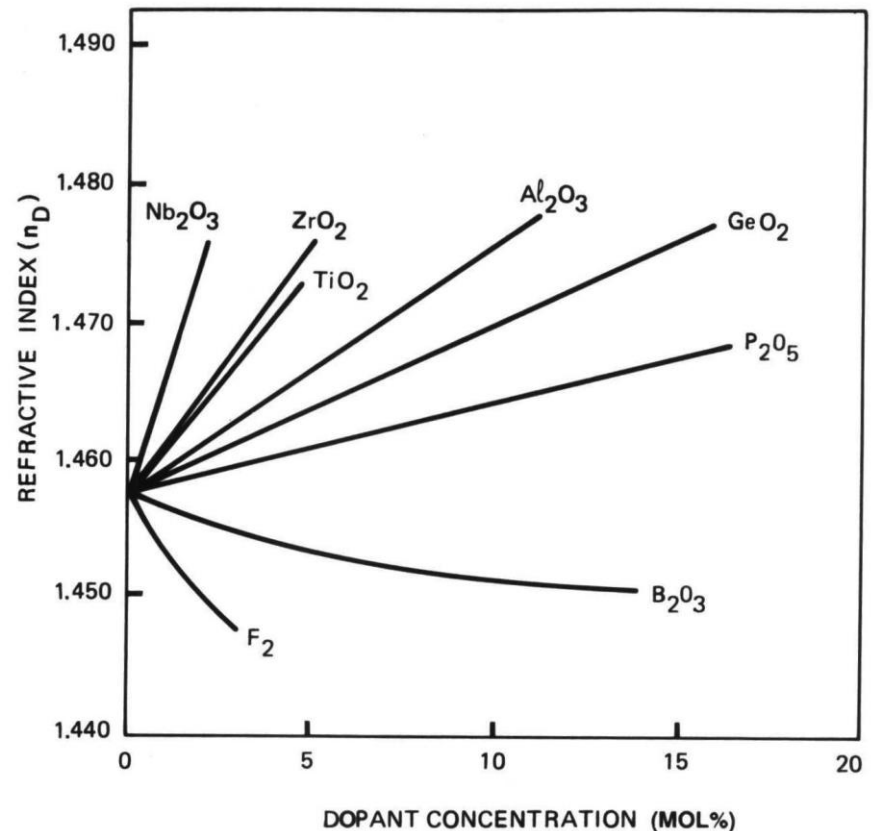
# Chemical vapor deposition

The refractive index profile is defined by doping silica in a controlled way. Different dopants can either increase or decrease the refractive index of the glass.

The most diffused dopant is germanium dioxide. It is obtained with a reaction similar to that used for silica, oxidizing germanium tetrachloride from the vapor phase:



Differently from the previous case, this reaction is reversible, posing some technical difficulties. Actually, the chloride produced by  $\text{SiCl}_4$  oxidation shift the equilibrium of the reaction toward left, hindering the oxidation of  $\text{GeCl}_4$ .

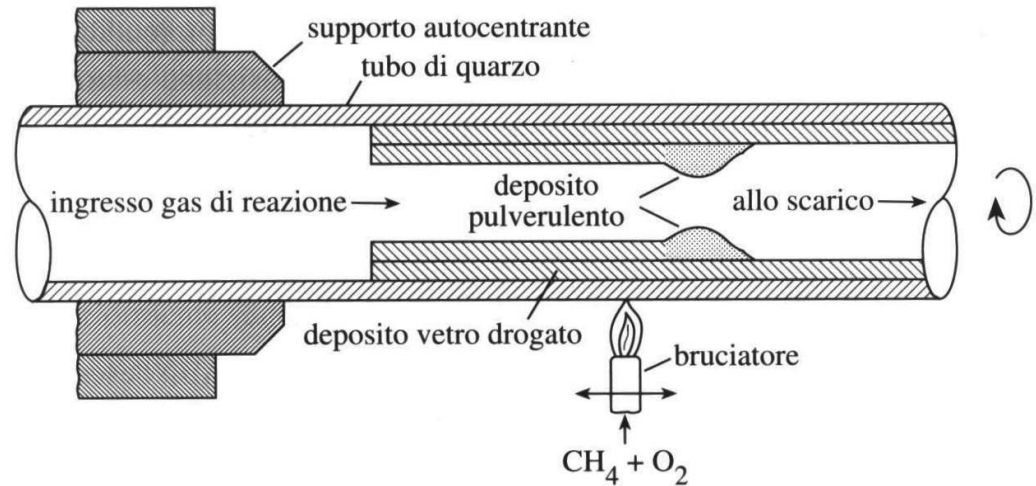


# Chemical vapor deposition

CVD can be implemented with different approaches, divided in to 2 main classes.

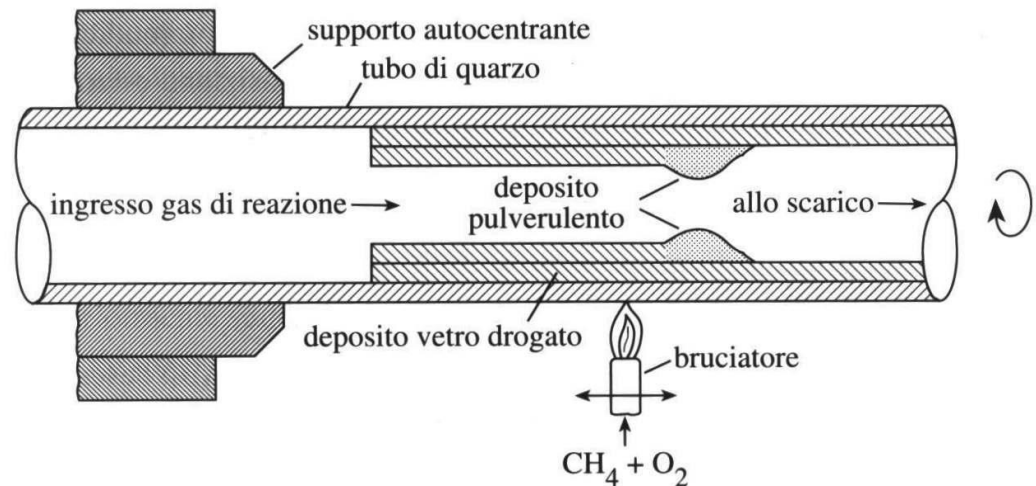
1. **IVPO** (inside vapour phase oxidation). In these class of processes the reaction occurs inside a cylinder of quartz that acts as support and is eventually incorporated into the preform. The following techniques belong to this class:
  - **MCVD**: modified chemical vapour deposition (Bell Labs, AT&T, 1974).
  - **PCVD**: plasma chemical vapour deposition (Philips, 1976).
2. **OVPO** (outside vapour phase oxidation), In these class of processes the reaction occurs outside a supporting bar that will not be part of the final preform. The following techniques belong to this class:
  - **OVD**: outside vapour deposition (Corning, 1972)
  - **VAD**: vapour axial deposition (NTT, 1977)

# MCVD



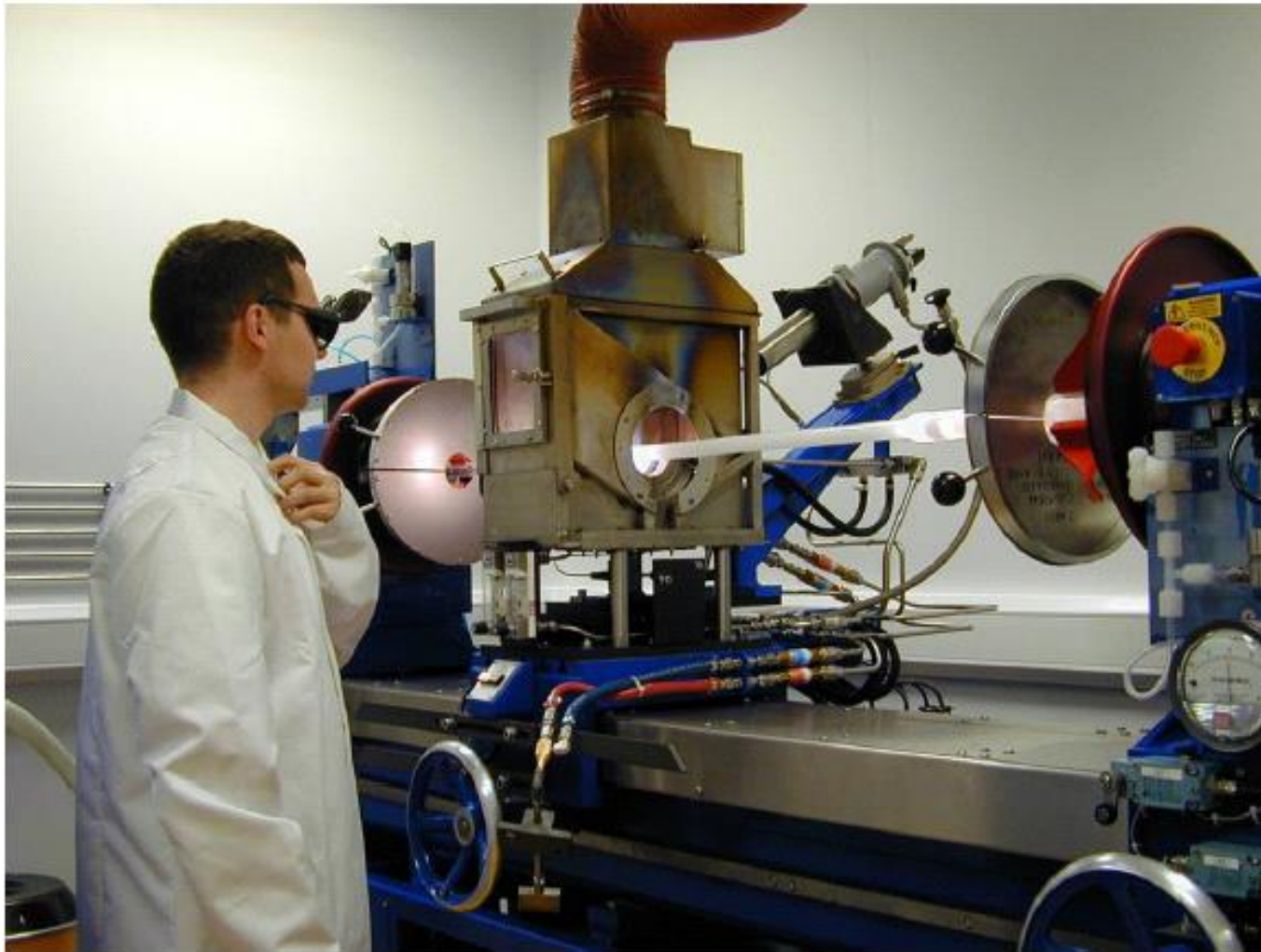
1. The reactant gasses ( $\text{SiCl}_4$ ,  $\text{GeCl}_4$ ,  $\text{O}_2$  plus other additives) move with laminar flow along a cylindrical tube of quartz, which is kept in rotation around its axis.
2. A burner travels along the tube at the same speed of the gasses; once at the tube end, it quickly goes back to the input and starts again.
3. The heat delivered by the burner increases the temperature above  $1200^\circ\text{C}$ , activating the oxidizing reaction and producing soot of silica or of any other dopant.

# MCVD



4. Due to the gas momentum and to the thermal gradient, the soot deposits on the tube inner wall in front of the burner, forming a porous layer.
5. As the burner proceed in its movement, it reaches the porous layer, heating and sintering it. In this way, a thin layer of silica glass (thickness between  $15\mu\text{m}$  and  $100\mu\text{m}$ ) is deposited at each passage of the burner.
6. When the central bore becomes small enough, the gas flow is stopped and the burner motion is slowed down to overheat the preform and collapse the central bore.

# MCVD



# MCVD

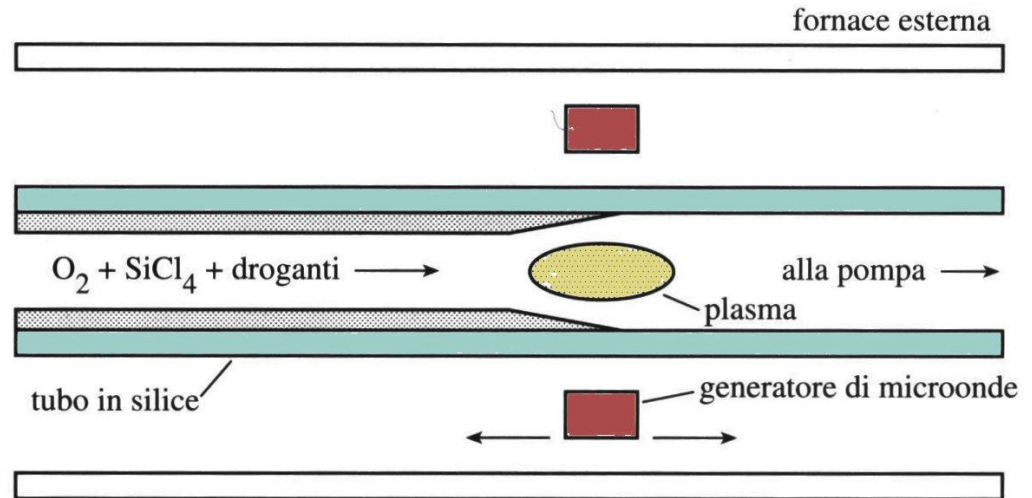
## Advantages:

1. Relatively simple and economic, enough to make it one of the most diffused techniques.

## Drawbacks:

1. Slow speed of deposition (between 0.5 g/min to 2 g/min).
2. Low efficiency: only 50% of the silica and 10-20% of germanium dioxide produced in the reaction deposit on the support.
3. The symmetry of the supporting cylinder critically impacts the quality of the final optical fiber.
4. As the central bore become smaller, the oxidation of  $\text{GeCl}_4$  is more and more hindered by the byproducts of  $\text{SiCl}_4$  oxidation. As a result, the refractive index profile typically shows a narrow deep at the fiber axis.
5. Any hydroxide ion ( $\text{OH}^-$ ) accidentally produced during the reaction will be permanently embedded in the preform, increasing the attenuation of the final optical fiber.

# PCVD



The working principle of PCVD is the same of MCVD, the only difference being the way in which heat is delivered to the reactants. In PCVD the burner is substituted by a microwave generator (at about 2.45 GHz) that induces a plasma in the reacting gasses, thus causing silica soot to form and precipitate.

An external heater keep the whole tube at high temperature to avoid thermal shocks.

# PCVD

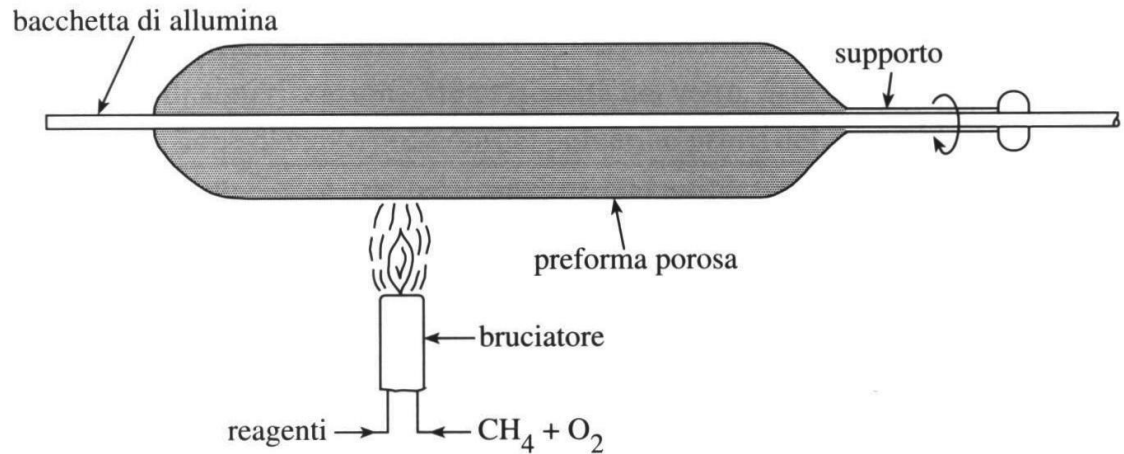
## Advantages:

1. Fairly high deposition speed (2-3 g/min).
2. High efficiency: 100% of the silica and 80% of the germanium dioxide deposit on the support.
3. Extremely high resolution in the definition of the refractive index profile; layers are about 1 $\mu$ m thick.

## Drawbacks:

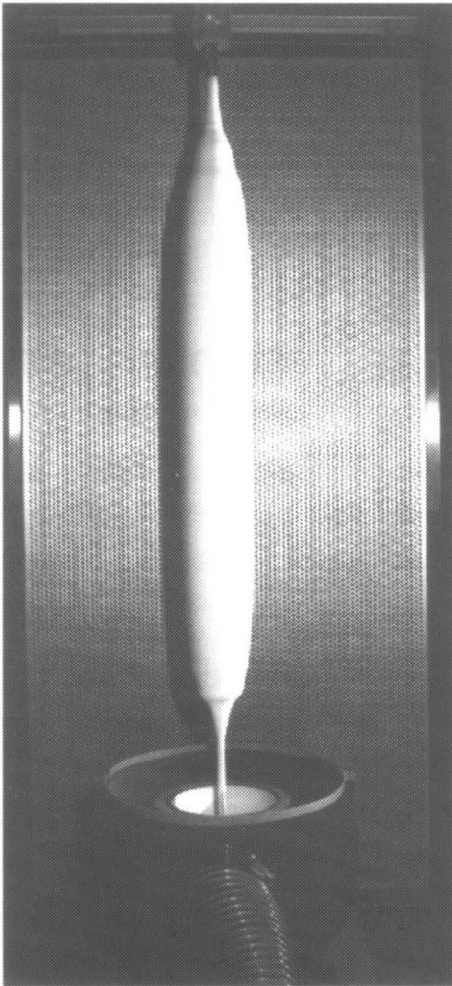
1. More expensive than MCVD.
2. The cylindrical symmetry of the final fiber depends on that of the support.
3. Any hydroxide ion ( $\text{OH}^-$ ) accidentally produced during the reaction will be permanently embedded in the preform, increasing the attenuation of the final optical fiber.

# OVD



1. A thin bar of refractive material (alumina or graphite) is kept in rotation and moved back and forth, while being exposed to the flame of a fixed burner.
2. A mix of reactants (SiCl<sub>4</sub>, GeCl<sub>4</sub>, plus other possible additive) and combustibles (methane and oxygen) are conveyed in the burner. Silica soot (diameter of about 0.1μm) forms in the flame.
3. Silica soot hits and adhere to the bar, growing a white porous prefom, with a density of about 15-20% that of vitreous silica (2.2 g/cm<sup>3</sup>). Each prefom layer of about 100μm thick.

# OVD



4. Once the preform is completed, the supporting bar is removed and the preform is inserted in furnaces at different temperatures and saturated with helium.
5. First, the preform undergoes drying at about 500°C. During this phase helium gas flows inside the porous structure mechanically removing OH<sup>-</sup> ions, largely present as a consequence of methane combustion.
6. Second, the temperature is increased up to about 1500°C to sinter the preform and to collapse the central hole.

# OVD

## Advantages:

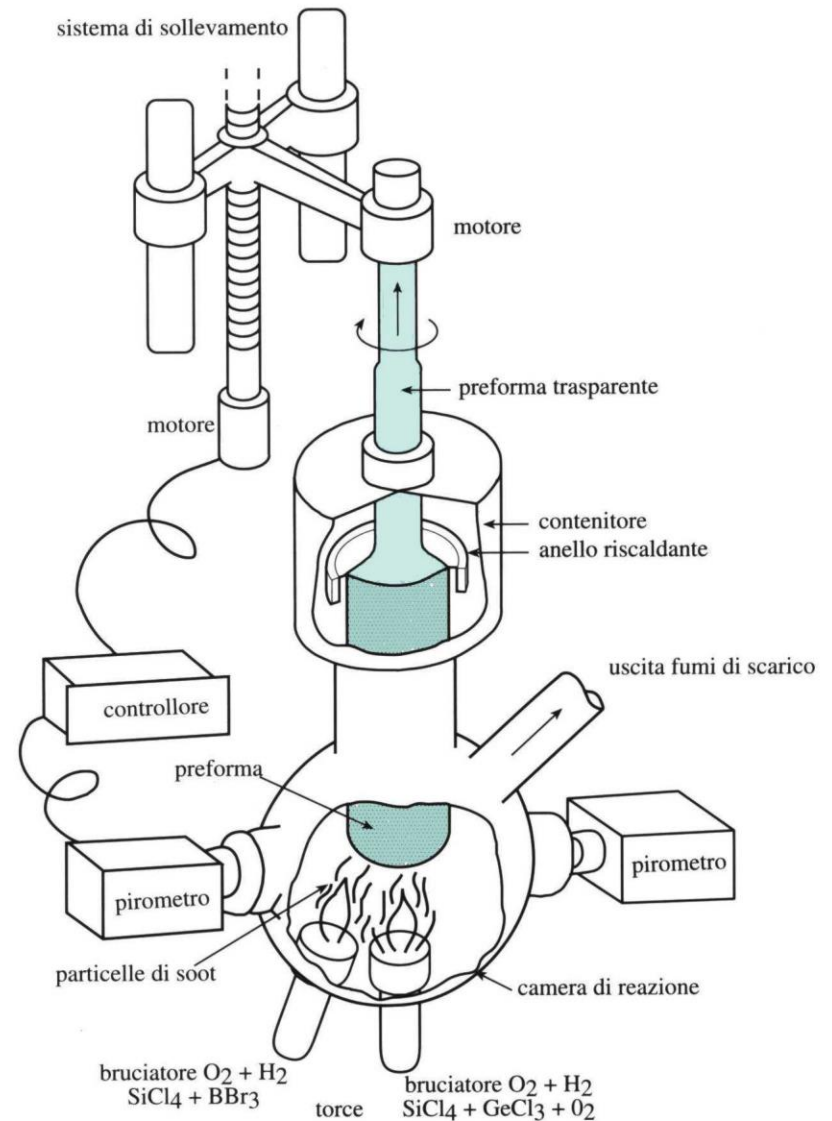
1. High speed of deposition (up to 10 g/min).
2. High efficiency. The water produced during methane combustion reacts with chlorine, forming hydrochloric acid (HCl) and oxygen; this subtracts chlorine from the environment easing the oxidation of  $\text{GeCl}_4$ .
3. The drying phase allows to drastically reduce contamination from  $\text{OH}^-$  ions.

## Drawbacks:

1. More difficult to implement and manage than MCVD.
2. The process must occur in clean rooms, to avoid contamination from the environment.

# VAD

1. A silica sphere is inserted in a reaction chamber, attached to a support that rotated and slowly moves upward. The sphere is exposed to the flames of a set of fixed burners.
2. As in the OVD process, reactants and combustibles are conveyed in the burners, so that the formed soot adheres to the sphere, growing a porous preform.
3. As the support moves upward, the preform enters other chambers where it is dried and then sintered.



# VAD

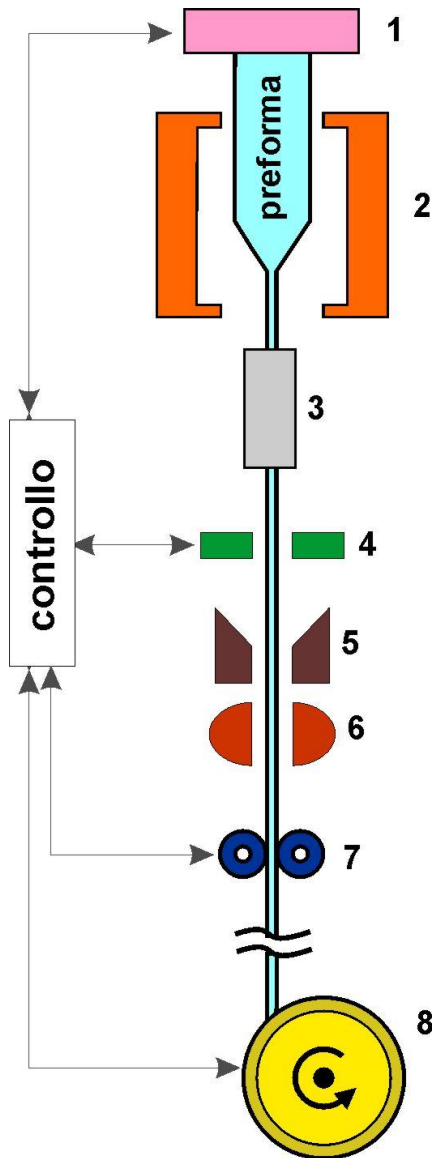
## Advantages:

1. High speed of deposition (up to 10 g/min).
2. High efficiency; as for OVD, the formation of hydrochloric acid eases the oxidation of  $\text{GeCl}_4$ .
3. The drying phase removes almost all the  $\text{OH}^-$  ions.
4. The production occurs "naturally" in a sealed and controlled environment.
5. The process allows the production of rather large preform, yielding up to 400 km of optical fiber each.

## Drawbacks:

1. More complex to implement and manage than MCVD.
2. The index profile cannot be easily changed, because it is set by the burner position of the fluidodynamic of the relative flames.

# Drawing

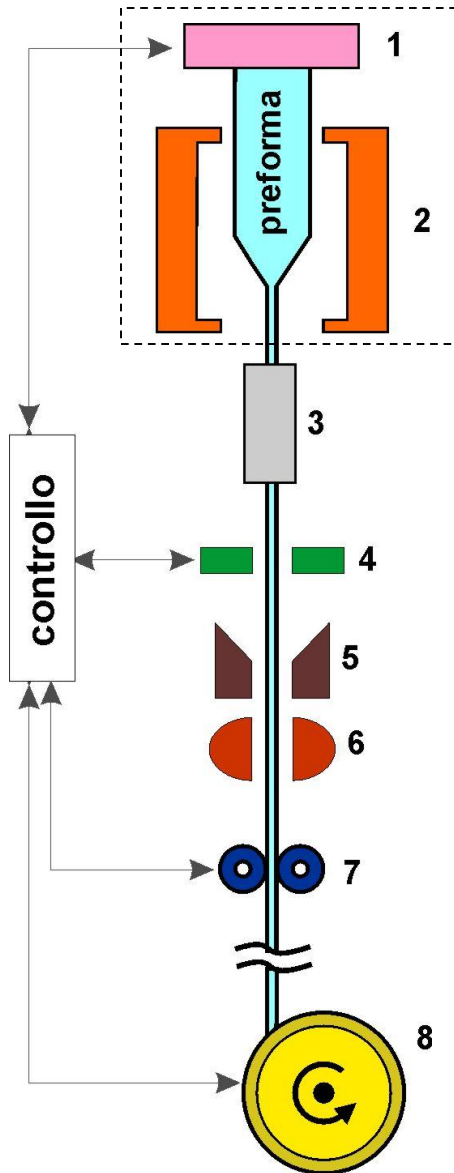


A preform is drawn into a fiber on specifically built **drawing towers**. The main components of these towers are the following:

1. Moving support to hang the preform.
2. Furnace.
3. Cooling tube.
4. Diameter monitoring.
5. Device to coat the fiber with resins.
6. UV lamps to polymerize the resins.
7. Spinner (optional)
8. Capstan.

The towers are typically from about 5 to 25 m tall. The taller the tower, the higher the production speed that can be achieved.

# Drawing

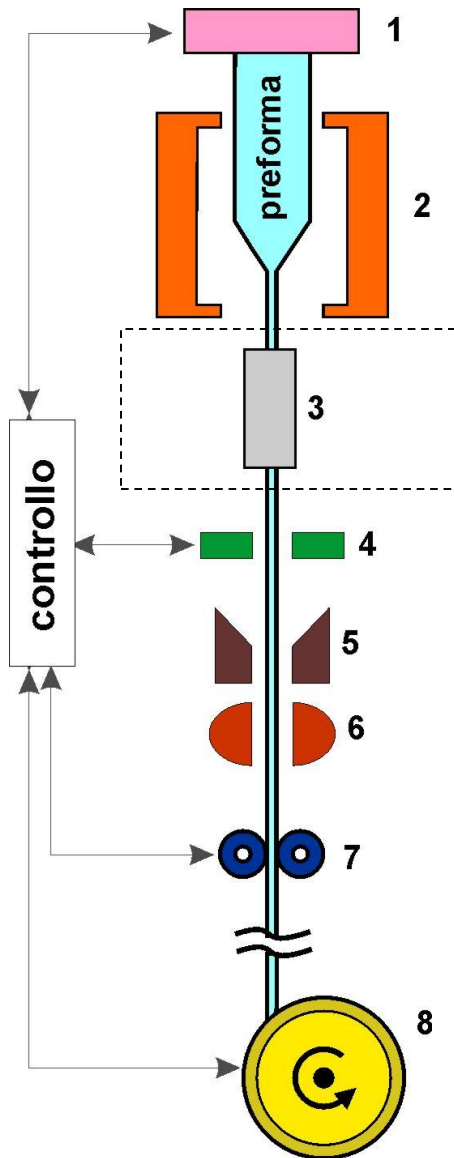


The preform is hung to a support (1) that slowly enters it in the furnace (2), where it is heated close to the silica melting temperature (about 2000 °C).

Modern furnaces are made of metal hollow cylinders heated by magnetic induction.

As the preform softens, a drop of silica start falling due to gravity, and the drawing process begins.

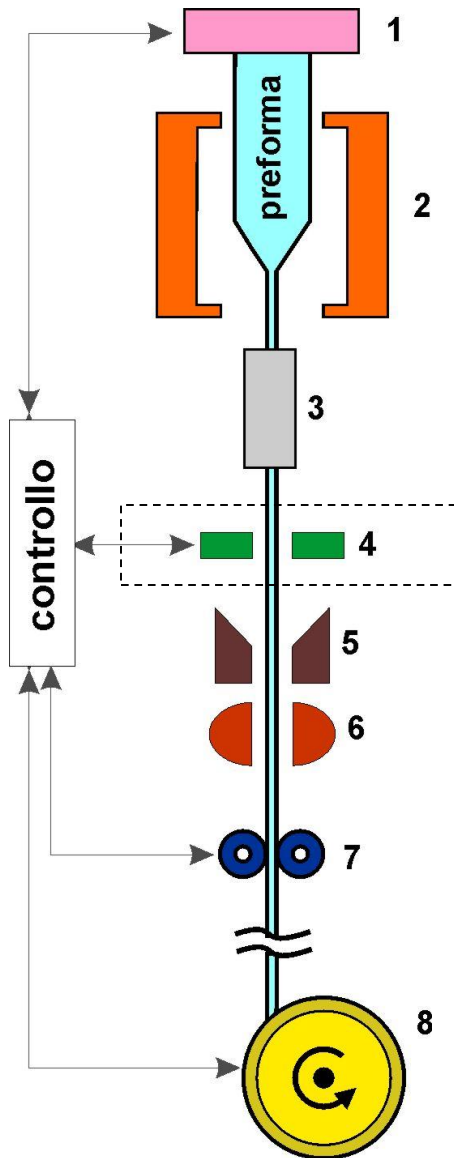
# Drawing



The drop and the connected fiber is guided by hand along the tower until the capstan (8). Afterwards, the drawing process continues under automatic control.

Out of the furnace, the fiber is first cooled down to handling temperature by passing it through cooling tubes (3), where it is exposed to flows of inert gases (such as helium).

# Drawing



One of the most critical parameters is the **fiber diameter**, which indirectly influences the propagation characteristic of the optical fiber. This is constantly monitored (possibly in different points along the tower) to keep it within design range.

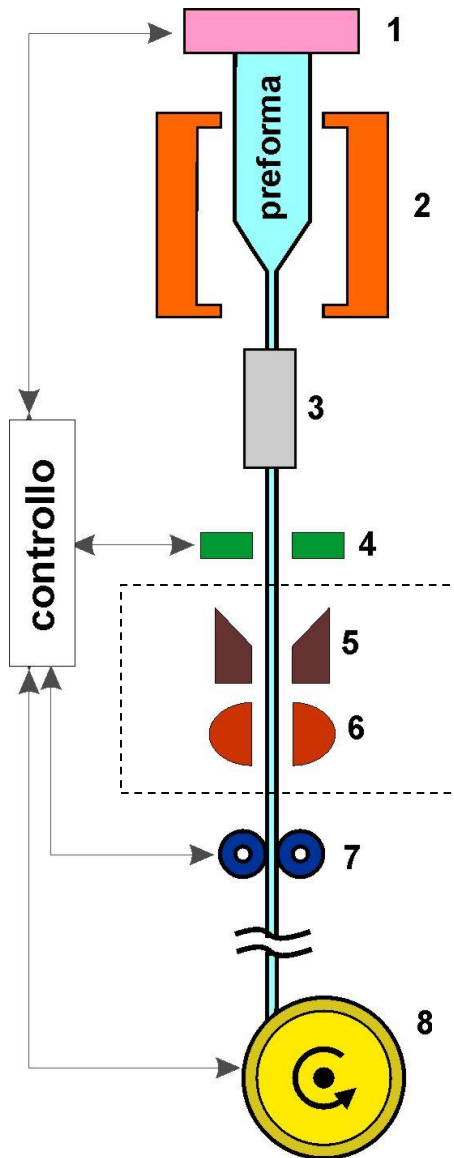
Let  $D_p$  be the preform diameter,  $d_f$  the fiber diameter,  $V_p$  the speed at which the preform enters the furnace and  $v_f$  the speed at which the fiber is drawn (controlled by the speed of rotation of the capstan); mass conservation implies that

$$D_p^2 V_p = d_f^2 v_f$$

Thus, fiber diameter can be controlled by the drawing speed.

Fiber diameter is measured by analyzing the interference patterns created by illuminating the fiber from its side.

# Drawing

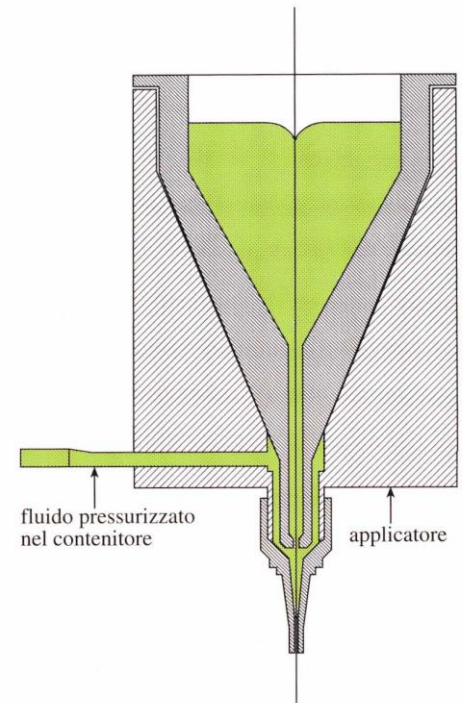


The cooled fiber is passed through a device (5) that coats it with acrylic resin; afterwards, the fiber is exposed to UV lamps and polymerize the resin. In this way the fiber is protected by the **primary coating**. The process can be repeated more times to apply more layers. Standard telecommunication fibers have two layers of coating.

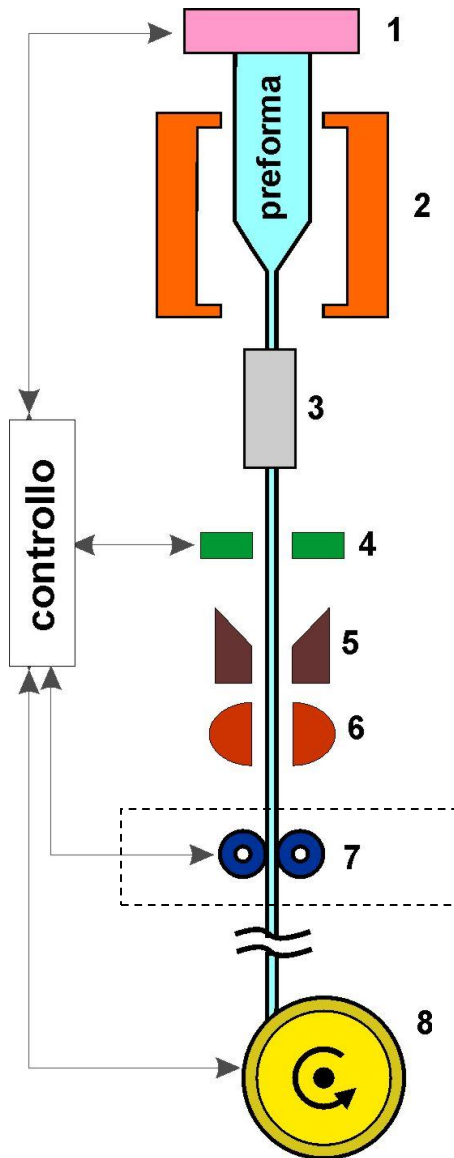
The first layer is rather soft (almost jelly); it provides mechanical isolation from the external environment and makes easy to remove the outer coating.

The second (outer) layer provides mechanical protection; it typically have low friction coefficient to easy cabling and winding of the fiber.

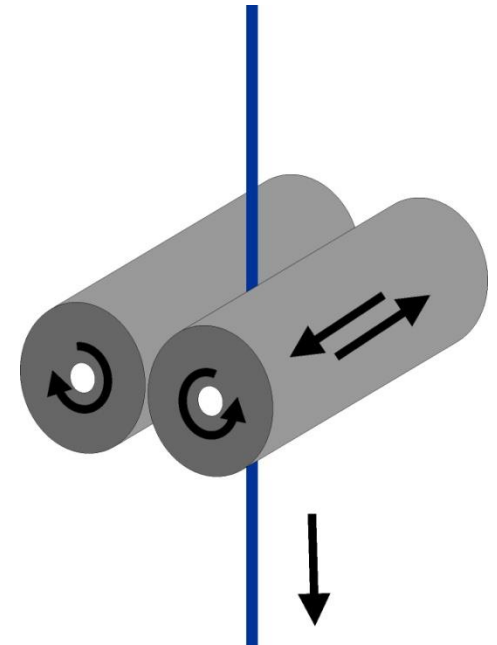
The final outer diameter is  $250\text{ }\mu\text{m}$  for standard for telecom fibers.



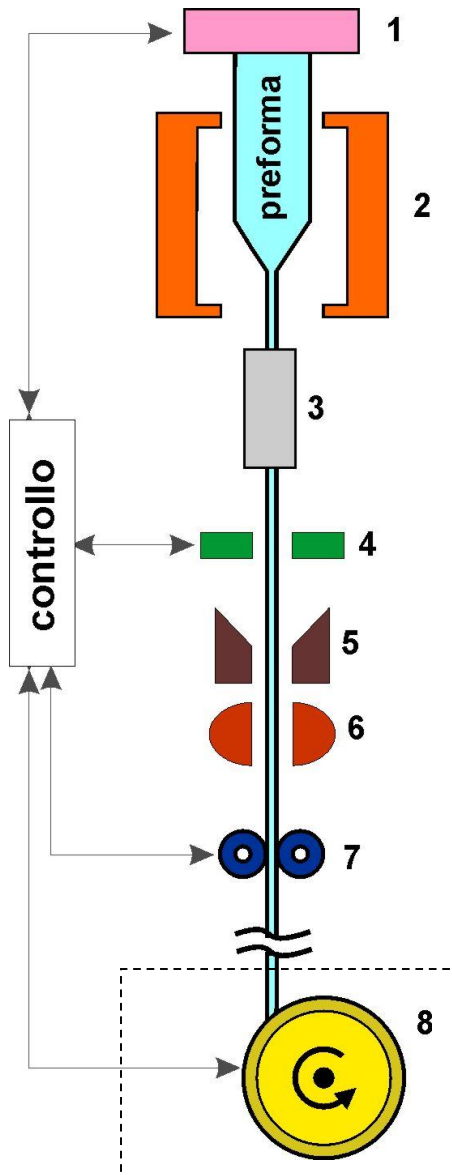
## Drawing



Sometimes, in particular for the production of high quality fibers, a **spinner** is inserted before the capstan. The spinner is a device able to induce a controlled rotation of the fiber around its axis. This allows to improve the cylindrical symmetry of the fiber, thus improving its transmission quality (reducing polarization mode dispersion). Spin rate can be as high as few hundreds of turns per second.



# Drawing



At the bottom of the tower the fiber is wound on the capstan, whose rotation speed controls the fiber diameter.

The tower is managed automatically by controlling furnace temperature, preform speed and capstan speed.

Drawing speed varies depending on the kind of fiber being drawn, but can be as high as several tens of meters per second for telecom fibers.

# Cabling

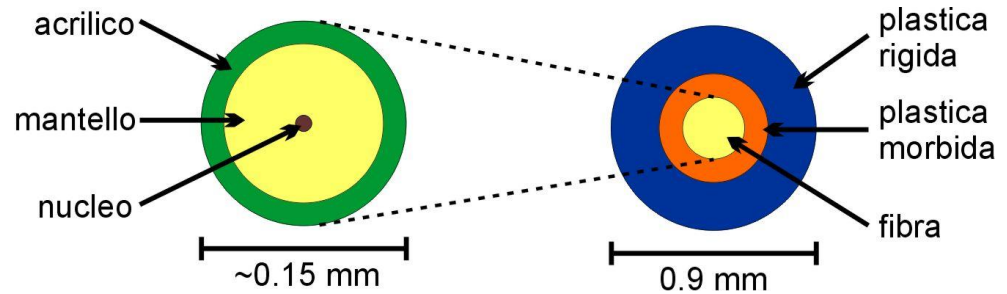
Despite a silica wire is stiffer than a steel wire with the same radius, silica is fragile and must be protected before installation. To this aim, different sheathings with different properties are applied, depending on the target use.

The production of optical fiber cables is not much different from the production of copper cables and shares most of the technological solutions.

# Primary protections

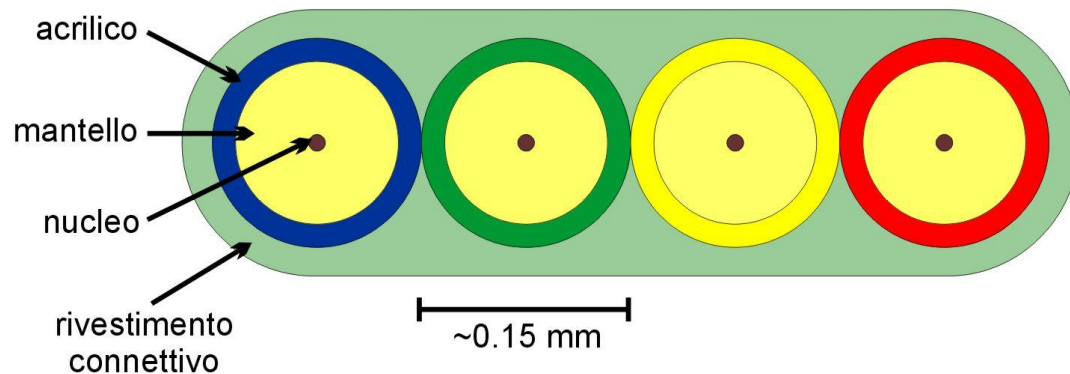
The **bare fiber** (i.e. the fiber protected only by the primary coating) is usually protected by the so called **900-micron buffer**, a layer of Teflon with a standard diameter of 900  $\mu\text{m}$ .

This kind of cable is mainly used in devices and in compound cables.



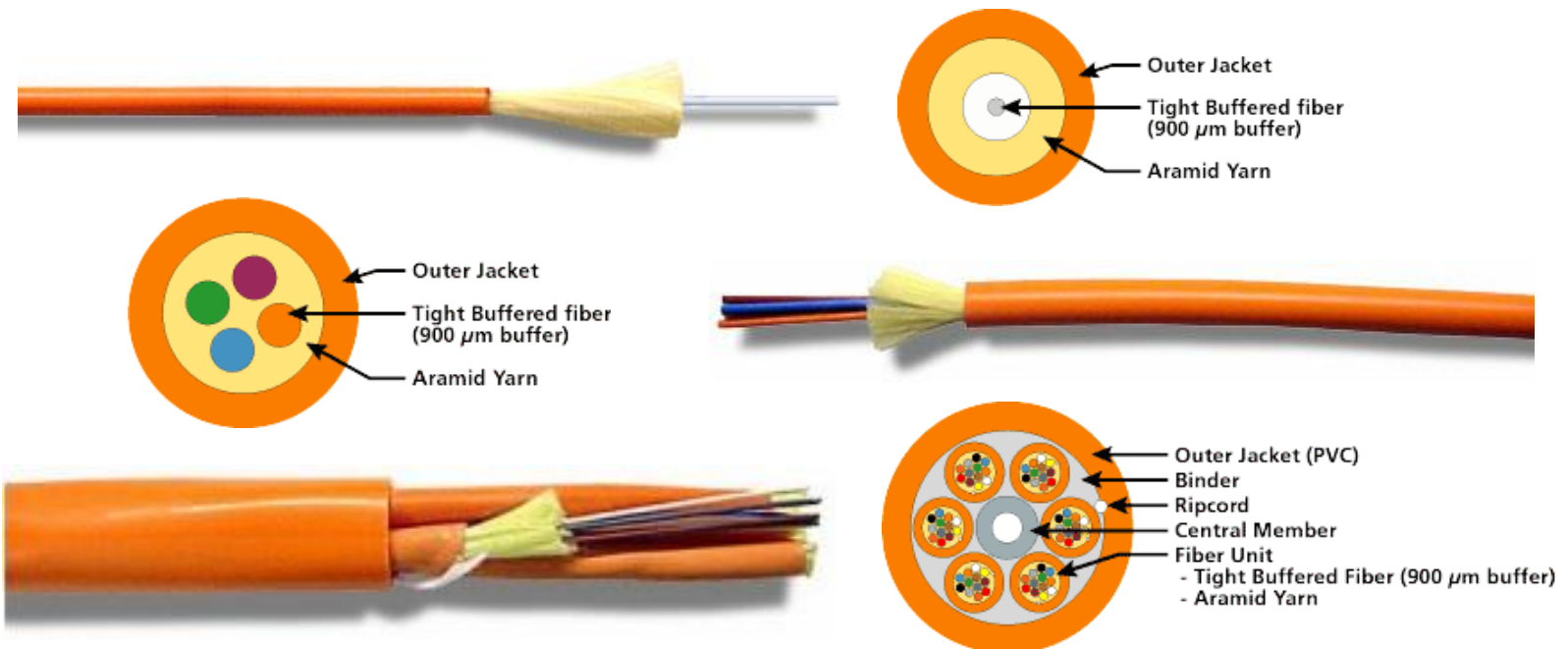
## Ribbons.

Bare fibers can be grouped in flat ribbons with 2, 4, 8 or 12 fibers. This structure simplifies the management and splicing of multi-fiber cables.

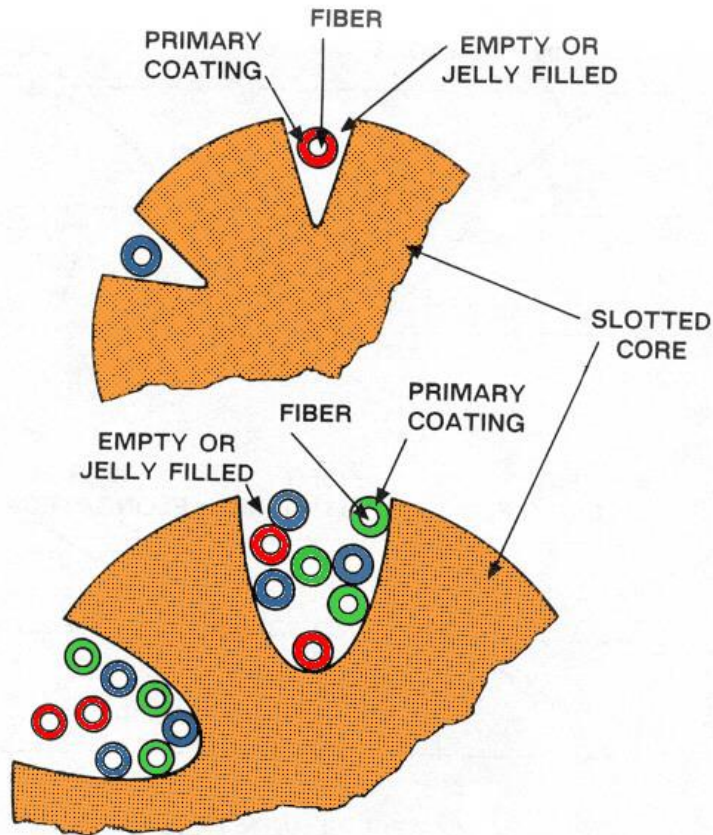


## Patch cords

Patch cords that will be installed in "safe" places are usually protected only by a sheathing that enclosed one or more 900-micron buffered fibers surrounded by a layer aramid yarn (usually Kevlar), whose aim is to withstand traction.

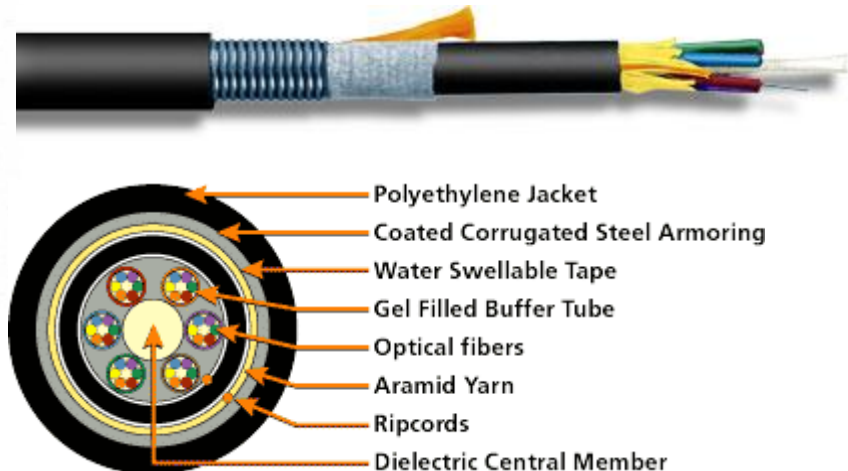


# Fiber optic cables

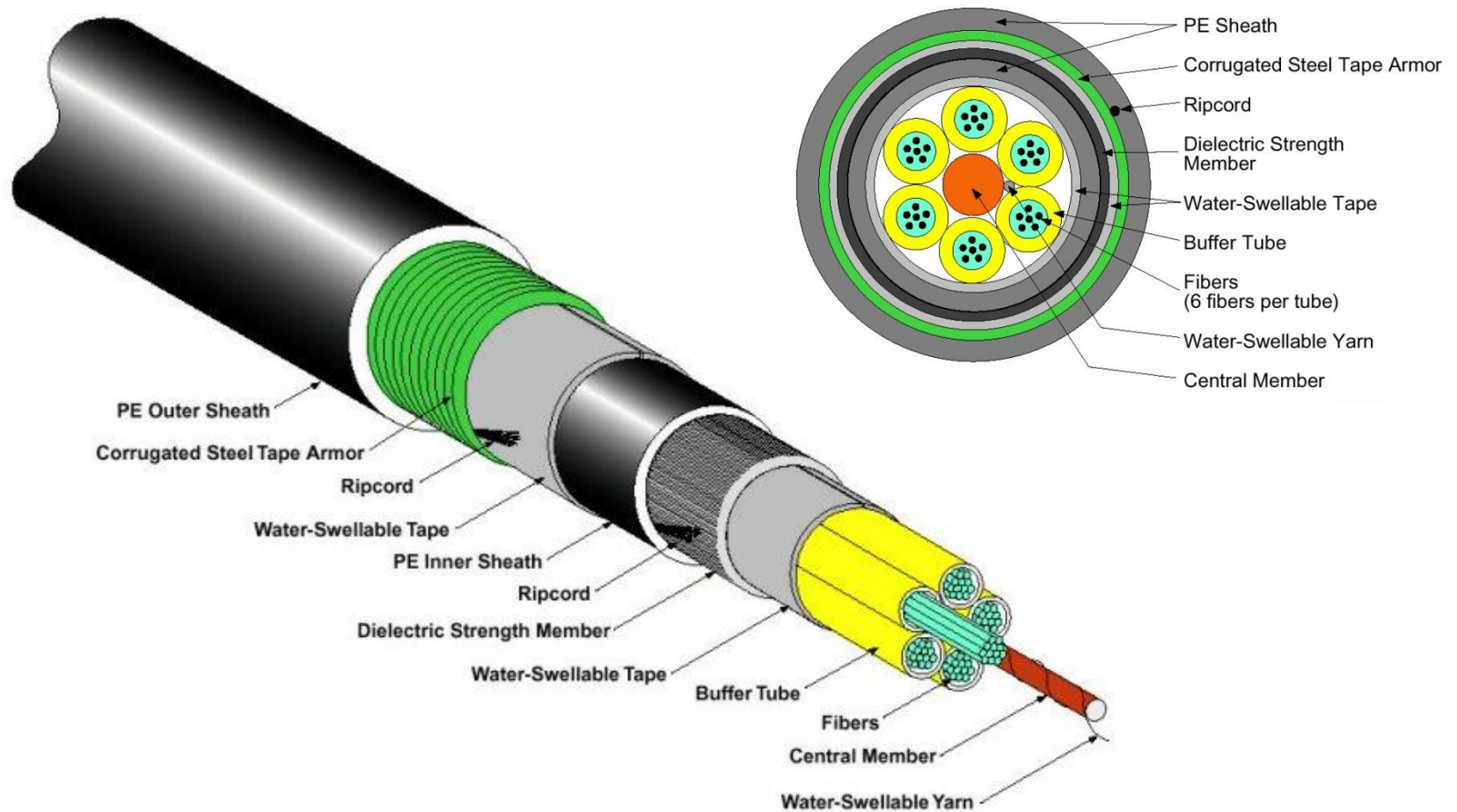


In multi-fiber cables the fibers are housed either in tubes wrapped around a supporting central dielectric rod, or in slots carved in the rod itself. central dielectric rod.

The central rod has also the fundamental role of contrasting thermal expansion (or contraction) of the cable.



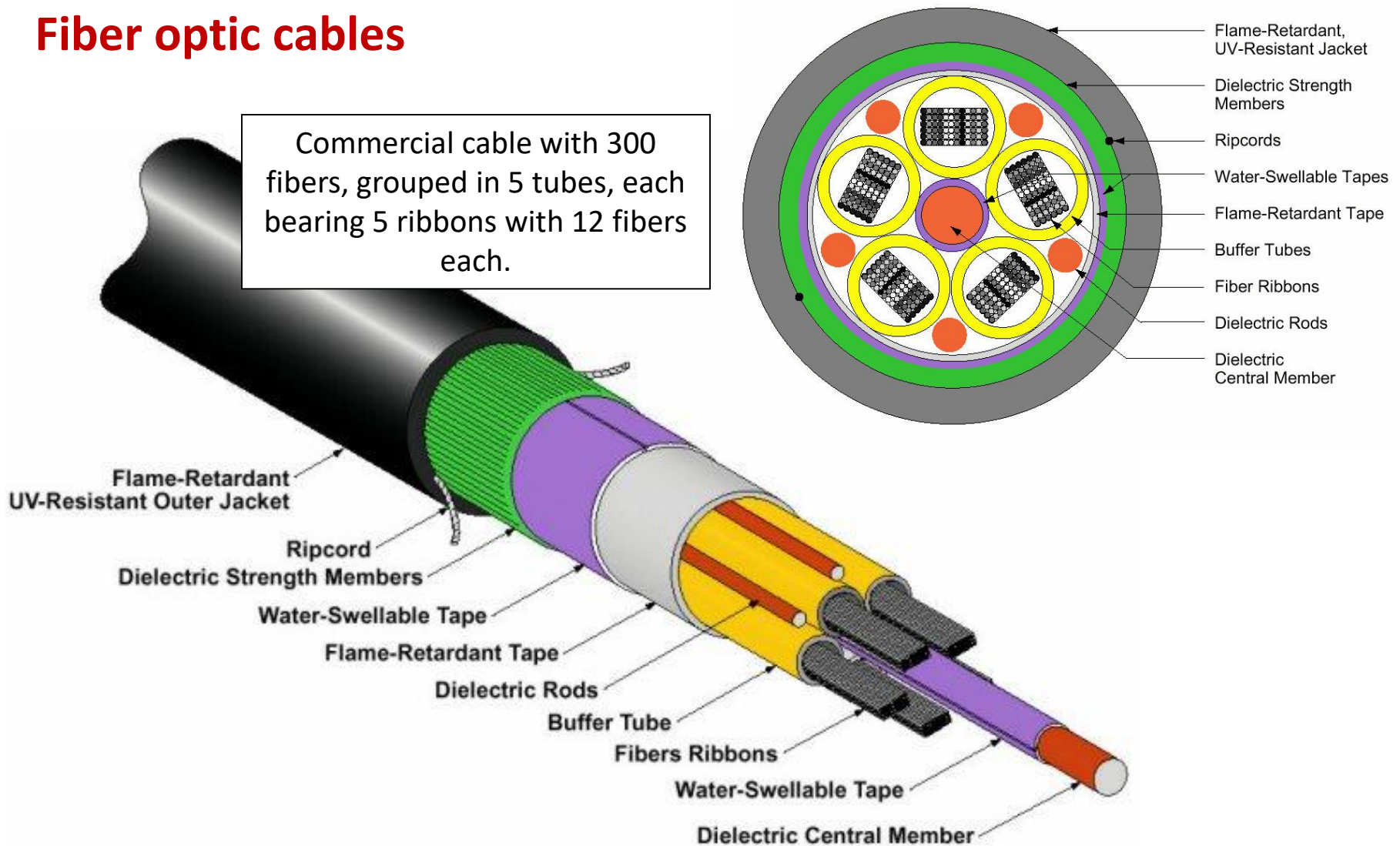
# Fiber optic cables



Cables have different protecting layers, with different functionalities. Among them there is often a layer of water-swellable tape, that expands when it come in contact with water, preventing water infiltration in case the external sheathing is damaged.

# Fiber optic cables

Commercial cable with 300 fibers, grouped in 5 tubes, each bearing 5 ribbons with 12 fibers each.



Metallic armoring is used to protect the cable from accidental hits and rodents.

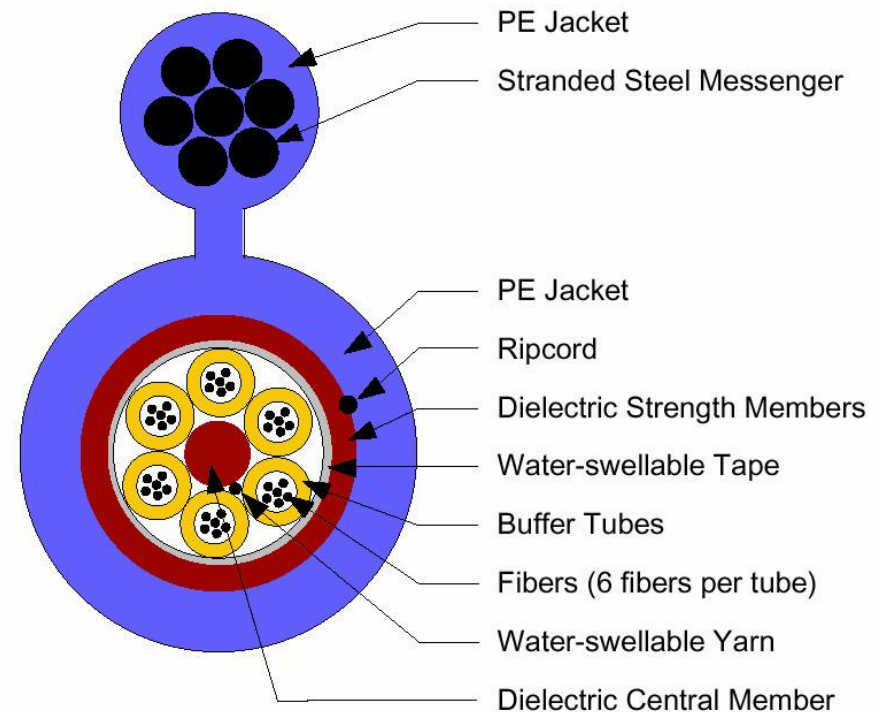
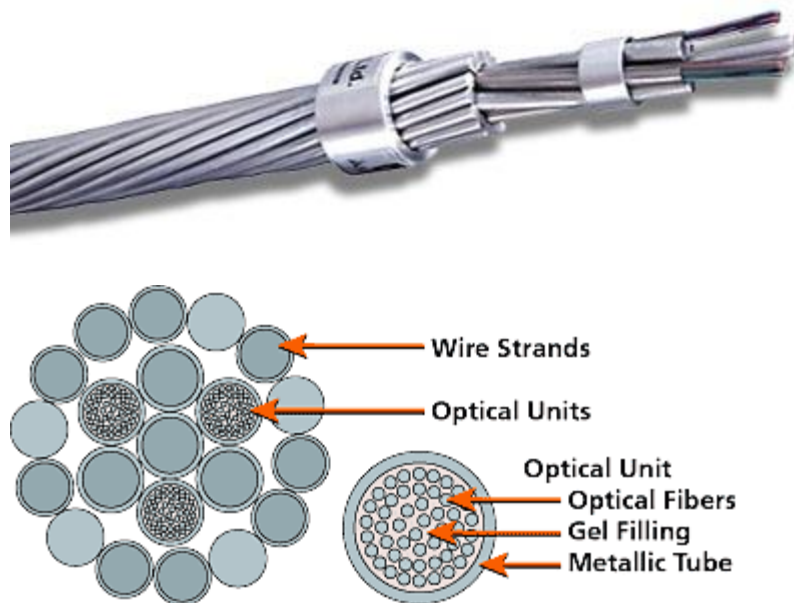
# Cable installation



Micro-trenching

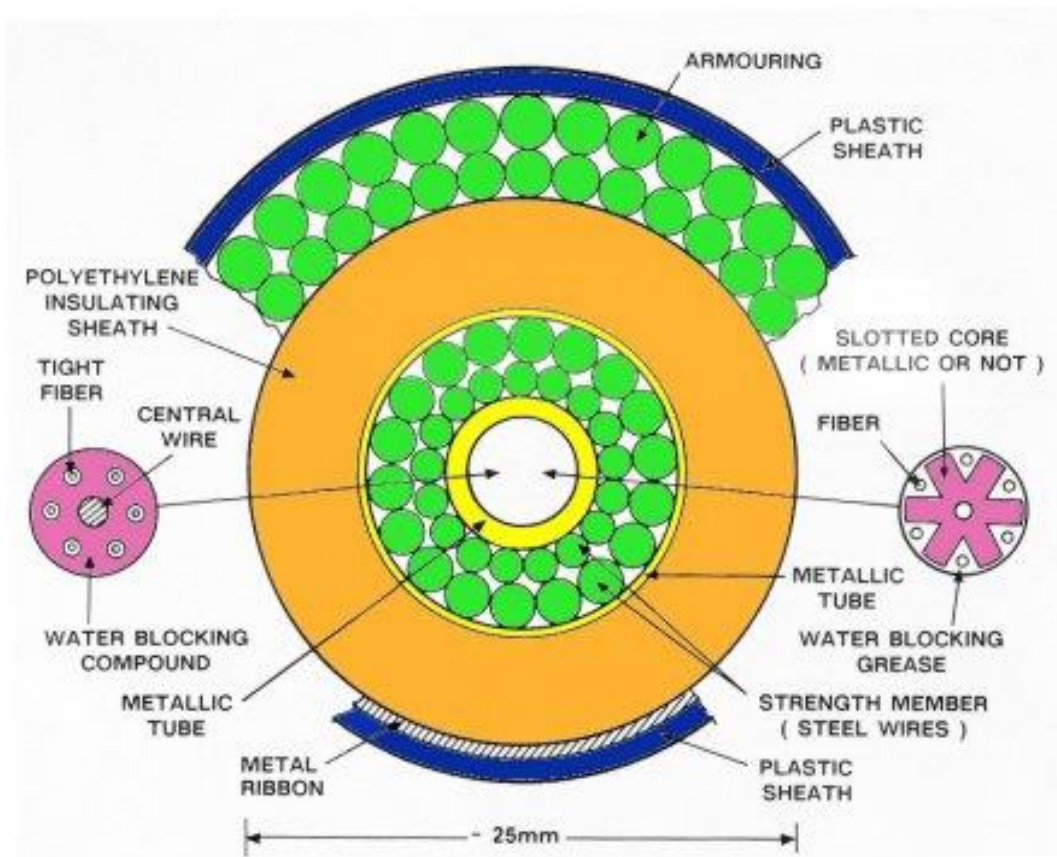
# Aerial cables

Metallic cables housing optical fibers can be installed in lieu of shield (or earth) wire, along high-voltage electric lines.



Aerial installations can be made also with self-bearing cables with a standardized shape, called **Figure-8**.

# Submarine cables



Submarine cables are heavily armored to withstand the installation process, the high pressure at the sea bed and accidental impacts from anchors and drag-nets. Close to the coast the cable protection is increased by reinforcing the armoring or burying the cable in the sea bed.

Submarine cables often include also electric wire to remotely power optical fiber amplifiers.

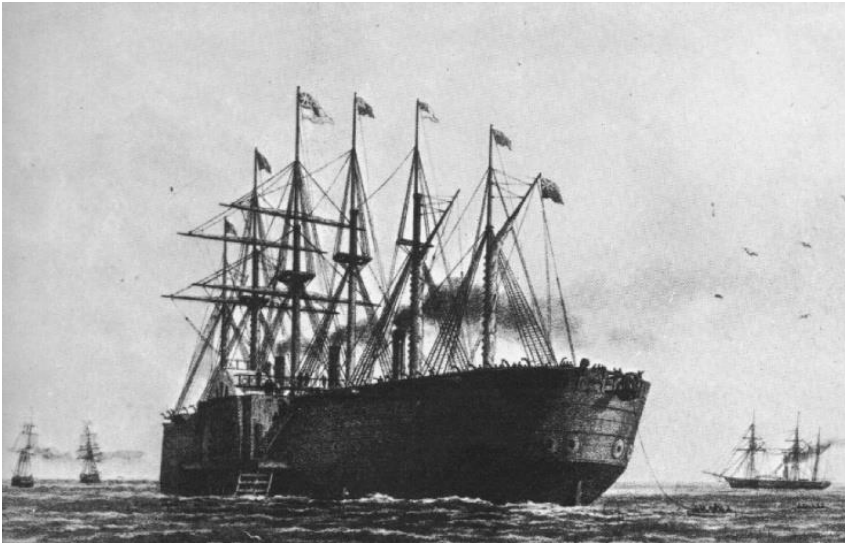
# Cable layer ships



The first submarine cable has been deployed between Canada and Ireland in 1858, by the ship Agamemnon, assisted by the Niagara. The link offered a capacity of 25 words per hour. Owing to the stress suffered during installation, the cable lasted only few weeks.

The first lasting installation was made in 1866 between London and New York; the link had a capacity of 8 words/min at a cost of about 5\$ per word. The installation was made by the **Great Eastern**.

The first ship entirely made of metal, the Great Eastern was 210 m in length and 5 times larger than the largest ship at that time.



It was however afflicted by several accidents, starting from the launching that took 3 months of efforts, until an exceptionally high tide made it possible. For this reason the Great Eastern was considered jinxed and during its maiden voyage it carried only 38 passengers. Among them there was Jules Verne, who took inspiration for the novel "A floating city".

The Great Eastern was design by the naval engineer **John Scott Russel**, who discovered also **soliton**, a particular class of waves that found some application also in optical fibers communication.

# Cable layer ships



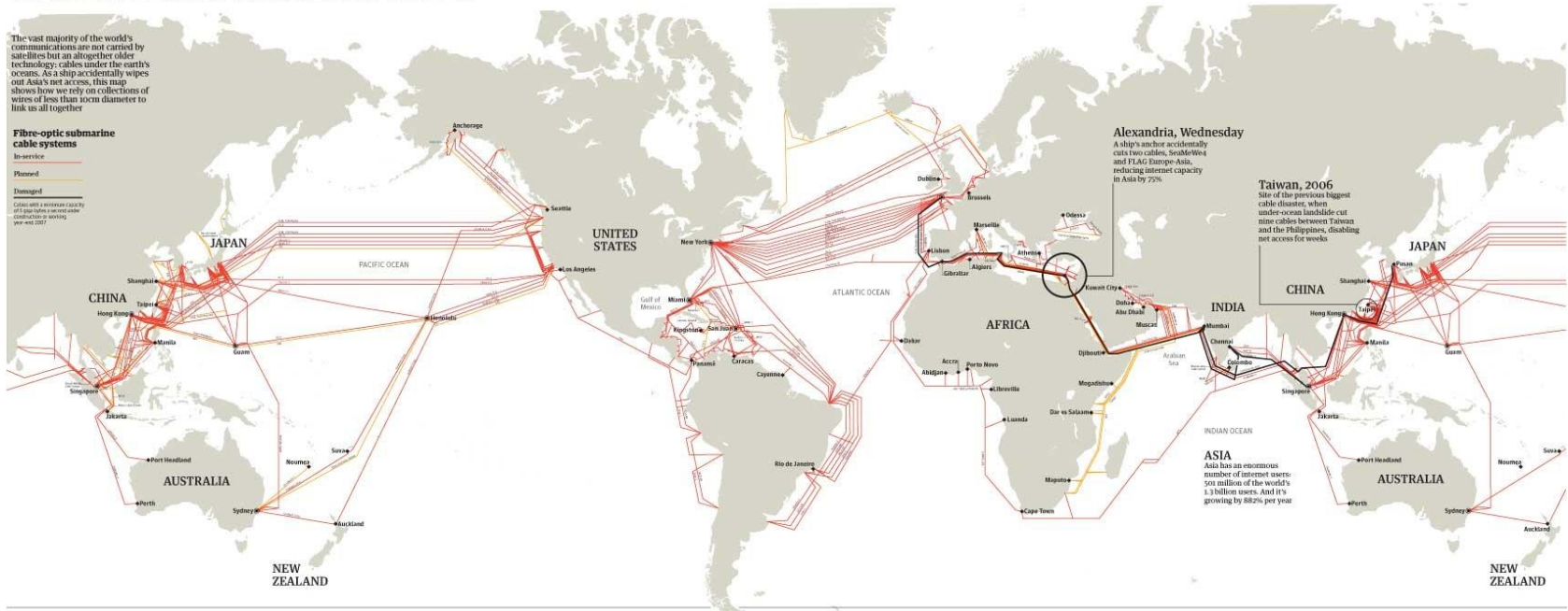
# Submarine cable network

## The internet's undersea world

The vast majority of the world's communications are not carried by satellites but an altogether older technology: cables under the earth's oceans. As a ship accidentally wipes out Asia's net access, this map shows how we rely on collections of wires of less than 1cm diameter to link us all together

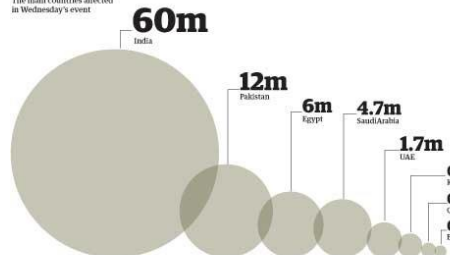
### Fibre-optic submarine cable systems

**In-service**  
**Planned**  
**Damaged**  
Capacity increased roughly 10% per year in recent years (2000-2007)



### Internet users affected by the Alexandria accident

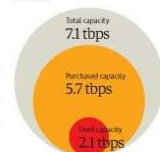
The main countries affected in Wednesday's event



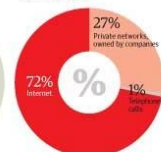
### World cable capacity

Submarine cable operators limit (turn on) capacity on their systems to sell bandwidth to other carriers. Carriers buy extra capacity, mainly to hold in reserve. On the trans-atlantic route 80% of the bandwidth is purchased, but only 29% is used

### Capacity in terabytes a second



### What makes up "used capacity"?



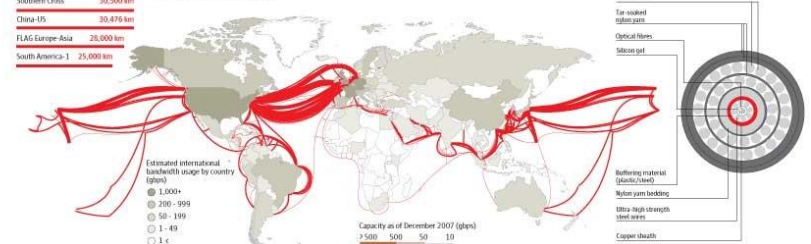
### The longest submarine cables

The Seasmith-3 system from Norden in Germany to Recife, South Korea connects 32 different countries with 39 landing points

|                  |           |
|------------------|-----------|
| Seasmith-3       | 39,000 km |
| Southern Cross   | 30,500 km |
| China-45         | 30,475 km |
| FLAG Europe-Asia | 28,000 km |
| South America-1  | 25,000 km |

### The world's cables in bandwidth

The first intercontinental telephony submarine cable system, TAT-1, connected North America to Europe in 1958 and had an initial capacity of 640,000 bytes per second. Since then, total trans-atlantic cable capacity has soared to over 7 trillion bps



<http://www.submarinecablemap.com/>